mechanical action of the wiper blades to wipe the outside surface of the glass, as well as into the hydraulic action of the washer system to apply washer fluid stored in an on-board reservoir to the area of the glass to be wiped. When combined, these components provide the means to effectively maintain clear visibility for the vehicle operator by removing excess accumulations of rain, snow, bugs, mud, or other minor debris from the outside windshield glass surface that might be encountered while driving the vehicle under numerous types of inclement operating conditions.

The vehicle operator initiates all front wiper and washer system functions with the control stalk and control knob of the right multi-function switch that extends from the right side of the steering column, just below the steering wheel. Moving the control stalk upward selects the front wiper system operating mode. The front wiper system allows the vehicle operator to select from two continuous wiper speeds, High or Low, or the intermittent wipe Delay mode. Pulling the control stalk downwards and releasing it selects the front wiper system Mist mode, which operates the front wipers for one complete wipe cycle after the control stalk is released. Rotating the control knob on the end of the control stalk allows the vehicle operator to select the intermittent wipe Delay interval. Pulling the control stalk towards the steering wheel activates the front washer pump/motor, which dispenses washer fluid onto the windshield glass through the front washer nozzle.

When the ignition switch is in the Accessory or On positions, battery current from a fuse in the Power Distribution Center (PDC) is provided to the front wiper and washer system fuse in the fuse block. This fuse provides battery current through separate fused ignition switch output (run-acc) circuits to the right multi-function switch, and to the front wiper motor. Within the right multi-function switch, this battery current is fed to one side of the control coil in the wiper motor relay, and to the electronic intermittent wipe logic circuit, which are both integral to the switch.

The intermittent wipe and wipe-after-wash features of the front wiper and washer system are both provided by the electronic intermittent wipe logic circuit within the right multi-function switch. In order to provide the intermittent wipe feature, the logic circuit monitors the wiper switch state, the intermittent delay resistance setting, and the wiper motor park switch state. In order to provide the wipe-after-wash feature, the logic circuit monitors both the front washer switch state and the wiper motor park switch state.

The hard wired circuits and components of the front wiper and washer system may be diagnosed and tested using conventional diagnostic tools and procedures. However, conventional diagnostic methods are not possible in the diagnosis of the intermittent wipe mode circuitry and logic within the right multi-function switch. If the front wiper and washer system operates satisfactorily in all modes except the Delay mode, the faulty right multi-function switch must be replaced. Refer to the appropriate diagnostic information.

OPERATING MODES

Following are paragraphs that briefly describe the operation of each of the front wiper and washer system operating modes.

CONTINUOUS WIPE MODE

When the Low position of the right multi-function switch control stalk is selected, the Low mode circuitry within the switch directs battery current to the low speed brush of the front wiper motor, which causes the front wipers to cycle at a low speed. When the High position of the control stalk is selected, the High mode circuitry within the switch directs battery current to the high speed brush of the front wiper motor, which causes the front wipers to cycle at a high speed.

When the Off position of the right multi-function switch control stalk is selected, one of two events is possible. The event that occurs depends upon the position of the wiper blades on the windshield at the moment that the control stalk Off position is selected. If the wiper blades are in the down position on the windshield when the Off position is selected, the park switch that is integral to the front wiper motor is closed to ground and the wiper motor ceases to operate.

If the wiper blades are not in the down position on the windshield at the moment the Off position is selected, the park switch is closed to battery current through the fused ignition switch output (run-acc) circuit of the front wiper motor. The park switch sense circuit directs this battery current to the low speed brush of the wiper motor through the normally closed circuit of the wiper motor relay and the Off mode circuitry of the wiper switch. This causes the wiper motor to continue running until the wiper blades are in the down position on the windshield and the park switch is again closed to ground.

INTERMITTENT WIPE MODE

When the control stalk of the right multi-function switch is moved to the Delay position, the Delay mode circuitry within the switch directs battery current to a request input of the integral logic circuit. The Delay mode circuitry also directs battery current through an internal variable resistor to the intermittent wipe delay sense input of the logic circuit, which indicates the delay interval that has been selected by the vehicle operator using the control knob on the end of the right multi-function switch control stalk.